Multiscale Semantic Fusion-Guided Fractal Convolutional Object Detection Network for Optical Remote Sensing Imagery

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***Abstract*— Optical remote sensing object detection is a challenging task, because of the complex background interference, ambiguous appearances of tiny objects, densely arranged circumstances, and multiclass object with vaster scale variances and irregular aspect ratios. The performance of object detection is seriously restricted. Thus, in this article, inspired by the anchor-free object detection framework, and aiming to solve these difficulties to improve the optical remote sensing object detection performance, a powerful one-stage detector of multiscale semantic fusion-guided fractal convolution network (MSFC-Net) is proposed. First, facing these strong-coupled semantic relations in each complex scene, a compound semantic feature fusion (CSFF) way is designed for generating an effective semantic description, which is a benefit to pixel-wise object center point interpretation. In addition, it can be easily extended into a semantic segmentation task. Second, in view of accurate multiclass pixel-wise center point predictions based on an effective compound semantic description, a novel fractal convolution (FC) regression layer is designed, which adaptively achieves the regression of multiscale bounding boxes (bboxes) with irregular aspect ratio under no priori information. Third, related to the set up FC regression layer, a specific hybrid loss is designed to make the proposed MSFC-Net converge better. Finally, the extensive experiments on challenge data sets of large-scale dataset for object detection in aerial images (DOTA) and object detection in optical remote sensing images (DIOR) datasets are carried out, and comparisons indicate that the proposed MSFC-Net can perform the remarkable performance than other state-of-the-art one-stage detectors, as it can reach 80.26% mean average precision (mAP) and 79.33% mF1 on DOTA and 70.08% mAP and 73.45% mF1 on DIOR. Then, our work is available at https://github.com/ZhAnGToNG1/MSFC-Net.**

***Index Terms*— Anchor-free, compound semantic description, fractal convolution (FC), multiscale feature fusion, object detection, optical remote sensing.**

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I. Introduction

OBJECT detection is an essential task for dealing with large-scale optical remote sensing imagery, and it can support military and civil intelligent surveillance systems for traffic control, urban planning, emergency rescue, and airport reconnaissance [1]–[9]. However, owing to usually encountered detection conditions including the tiny object with ambiguous appearances or disturbed by complex background (BG) interference, densely arranged circumstance and multiclass object detection with the vaster scale variances and irregular aspect ratios, the performance of object detection from optical remote sensing imagery is seriously restricted by these difficulties. Thus, for improving the performance, there are a lot of efforts have been made [10]–[52], [73]–[85], and these methods can be divided into two categories: 1) traditional hand craft feature-based methods [10]–[20] and 2) convolution neural networks (CNNs)-based one- or two-stage detectors [21]–[52], [73]–[85].

First, since hand-craft features' (e.g., histogram of orientation gradient (HOG) [10], scale-invariant feature transform (SIFT) [11], Hough transform [14], bag of words (BOW) [16], and local binary pattern (LBP) [17]) generalization ability is very limited for multiclass and -scale object detection from optical remote sensing scenes, traditional methods often provide poor detection performance. Recently, with the development of CNNs [53]–[58], there are a lot of famous one- or two-stage detectors (e.g., Faster-region-CNN (RCNN) [59], Yolo series v1-v5 [60]–[64], path aggregation network (PANet) [65], RetinaNet [66], receptive field block network (RFB-Net) [67], EfficientDet [68], CenterNet [69], fully convolutional one-stage (FCOS) [70], CornerNet [71], and FoveaBox [72]), which have achieved a great success and widely migrated into optical remote sensing object detection. For example, Van Etten [25] proposed a one-stage detector named you only look twice (YOLT) for multiscale object detection based on the Yolo framework, in which a fine-grained regression layer is used for tiny or medium object detection and a coarse-grained regression layer is used for large-scale object detection. Referring to soft non-maximum suppression (NMS), Dong *et al.* [28] proposed a novel Sig-NMS method for Faster-RCNN to rescore the detected anchors by a sigmoid function and ensure tiny and

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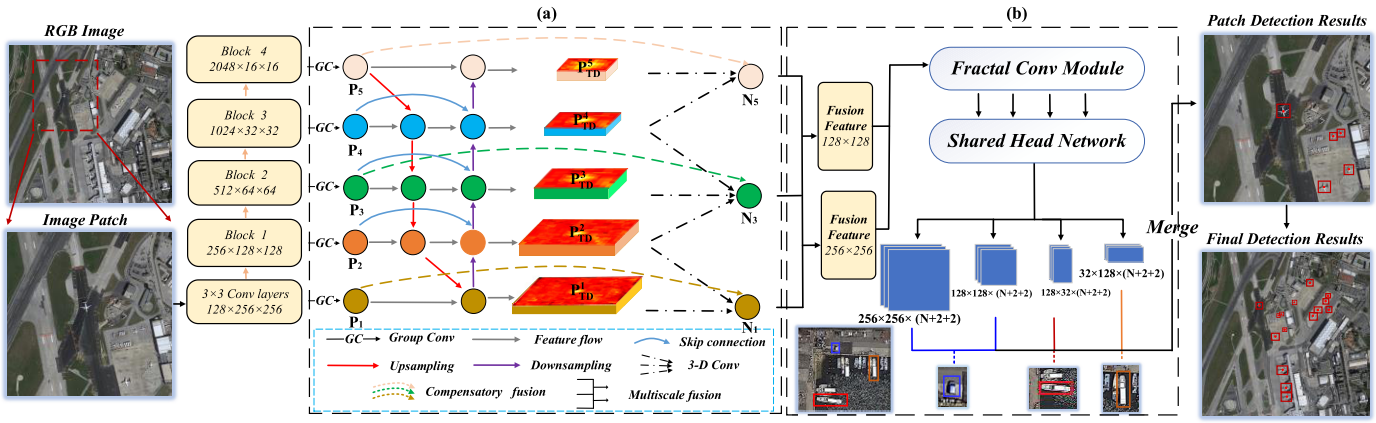


Fig. 1. Architecture of the one-stage anchor-free MSFC-Net. (a) CSFF way. (b) FC regression layers.

densely arranged object detection scenarios. Dong *et al.* [33] utilized the K-means algorithm to initialize suitable anchor scales for Faster-RCNN to achieve better optical remote sensing object detection results. Based on the Yolo-v3 framework, Qin *et al.* [35] proposed an effective feature fusion strategy to make full use of the feature representations at different scales while utilizing the parallel-dilated convolutions to obtain additional context information to support tiny object detection. Moreover, Jin and Lin [38] expected the maximum overlapping Intersection over Unions (IoUs) between ground truths (GTs) and predicted bounding boxes (bboxes) to set up an anchor selection module that could reduce the redundancy of initial anchors for model training and achieve the multiscale object detection in aerial images. As mentioned above, although a lot of anchor-based detectors are designed to solve these difficulties in optical remote sensing object detection, they cannot provide satisfactory results which are caused by the limitation about detection mechanism of fixed anchor region localization and classification. Specifically, the discrete initial anchor setting is hard to describe the continuous scale variances within a large scope and irregular aspect ratios [52]. Then, especially for tiny object detection, the localization and classification limited in fixed anchor regions would be severely disturbed by the ambiguous appearance and unseen interference. In addition, related to the postprocessing of NMS in anchor-based methods, it also would take a big risk to arise the missing detection from densely arranged circumstances.

Thus, in this article, we rethink these difficulties in optical remote sensing object detection and are inspired by anchor-free detectors (e.g., CenterNet [69], FCOS [70], and CornerNet [71]) utilizing keypoint interpretation for object detection. A new one-stage anchor-free detector called multiscale semantic fusion-guided fractal convolution network (MSFC-Net) is set up, as shown in Fig. 1 and it is very suitable for optical remote sensing object detection. First, we consider establishing the coupled or uncoupled semantic relations in complex optical remote sensing scenes. Consequently, as shown in Fig. 1(a), a compound semantic feature fusion (CSFF) way is designed for constructing an effective semantic description and achieving accurate pixel-wise object center point prediction. Here, object center points are chosen as keypoint interpretation and the reason is that center points never overlap according to overlooking observation so that it is friendly for densely arranged object detection. In addition, object center points can be interpreted by encoder-decoder structure in CNN referring to semantic segmentation technology [86]–[90], especially for tiny object detection, because the more adequate semantic relation among different objects and unseen BG interference can effectively improve their detection performance and avoid false alarms from complex BG, simultaneously. After the CSFF stage, accurate object center points can be predicted, however, in an anchor-free manner, the detector does not have any priori information of anchor scales and aspect ratios. Thus, based on center point prediction, how to generate the bounding boxes (bboxes) without any priori information becomes a challenging task. Therefore, we regard continuous scale and irregular aspect ratio changes of bboxes as a global and local self-similar fractal system. Meanwhile, considering that the center point prediction benefits from the same sample rates of feature extraction along vertical and horizontal directions, a fractal convolution (FC) regression layer is set up as shown in Fig. 1(b), which can achieve the regression of multiscale bboxes with irregular aspect ratios and generate accurate detection results without any priori information. Besides, corresponding to a set up FC regression layer, a specific hybrid loss is designed for boosting the convergence of the model in the training phase, which can achieve scale-invariant regression by introducing size constraint term and generalized IoU (GIoU) [107]. Finally, extensive experiments are carried out on challenge data sets of large-scale dataset for object detection in aerial images (DOTA) [98] and object detection in optical remote sensing images (DIOR) [99], and comparison results demonstrate that the proposed MSFC-Net can achieve superior detection performance than those of state-of-the-art one-stage detectors; it can reach 80.26% mean average precision (mAP) and 79.33% mF1 on DOTA [98] and 70.08% mAP and 73.45% mF1 on DIOR [99], respectively. In general, the contributions of our work can be summarized as the four points below:

1) A novel encoder-decoder architecture with a CSFF way is designed for effective semantic description in anchor-free detectors. It can set up the coupled or uncoupled semantic relations for each kind of pixel, which is a benefit to accurate object center point

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prediction and meanwhile avoid false alarms from complex BG. In addition, the designed CSFF way can be easily extended into semantic segmentation for land use classification, which also can achieve state-of-the-art performance comparing with other encoder-decoder ways.

2) Considering the continuous scale variance and irregular aspect ratio regression of multiclass object bboxes, a new FC regression layer is proposed under the anchor-free paradigm to adaptively fit the large scope scale variance and irregular aspect ratio without any priori information, and it also can ensure the same sampling rates of feature extraction along the vertical and horizontal directions, which can further facilitate object center point prediction and refined bboxes generation under extreme aspect ratio case.
3) In order to make the proposed MSFC-Net converge reasonably and quickly, a hybrid loss is designed, which not only considers the localization performance of object center point, but also introduces the scale-invariant constraint of GIoU [107] to treat the different scale objects equally in the training phase. By utilizing our designed hybrid loss, the proposed MSFC-Net can generate more accurate bboxes whether for tiny or large scale objects.
4) The compound semantic description and the fractal system principle are first introduced into optical remote sensing object detection, which can make the proposed MSFC-Net obtain superior detection performance than the state-of-the-art one-stage detectors.

The rest of this article is organized as follows. Section II briefly reviews the related works. Section III introduces the proposed MSFC-Net in three parts: 1) CSFF; 2) FC regression layer; and 3) hybrid loss for model training. In addition, comprehensive experiments and comparisons are reported in Section IV. Then, Section V provides the conclusion of this article.

II. Related Work and Problem Analysis

In this section, first, recently proposed anchor-free detectors applied for optical remote sensing scenes are discussed. Next, we individually analyze the several related works of pixel-wise keypoint interpretation and multiscale regression layer with hybrid loss.

A. Anchor-Free Object Detectors Applied for Optical Remote Sensing Object Detection

Recently, anchor-free object detectors (e.g., CenterNet [69] FCOS [70] and CornerNet [71]) have exhibited outstanding detection performance in natural scenes, and these methods break the restrictions of anchor-based detectors and achieve more concise and accurate object detection results. In view of these contributions, many studies about anchor-free detectors have been applied to optical remote sensing object detection [73]–[85], which can be summarized as two categories including the keypoint group-based [73]–[79] and center point-based [80]–[85]. First, related to keypoint group-based methods, Wei *et al.* [73] proposed an anchor-free X-LineNet for airplane detection, and this model can learn the rich compositional visual grammar information of airplanes' keypoints and pairing of intersecting line segments. Zhou *et al.* [74] also proposed a novel anchor-free airplane detector based on keypoint group prediction including head, left wing, body, right wing, and tail of airplanes. In addition, Chen *et al.* [75] designed an anchor-free detector for ship detection via keypoint group prediction (e.g., bow, center, and stern), which was a similar idea to that in [74]. Next, Zhang *et al.* [84] set up an anchor-free detector called Gaussian-mask regions network (GRS-Net) for rotation ship detection, and they can achieve rotated ship region detection by keypoint group prediction which is based on the constraints from Gaussian-mask region, kernel, geometry, and angle maps. Second, for center point-based methods, Fu *et al.* [81] designed a feature balancing and refinement anchor-free detector for multiscale ship detection from synthetic aperture radar (SAR) images, which is based on accurate center point prediction and multiscale bboxes regression. Shi *et al.* [82] proposed a centerness-aware network called CANet for object detection in remote sensing images. This method is focused on optimizing the pixel-level center point semantic description in each scale level of the feature pyramid network (FPN). Wang *et al.* [83] introduced a pixel-wise center map prediction branch to learn the center probability map and leverage oriented object detection in aerial images. Moreover, Zhang *et al.* [84] set up an anchor-free detector for arbitrarily oriented ship detection based on accurate center point predictions. In general, these anchor-free detectors mentioned above all have to construct an effective multiscale semantic description for keypoint group or center point interpretation. However, the keypoint group-based detectors mainly focus on dealing with a specific object (i.e., airplanes or ships) that commonly have a large or medium scale and very clear appearances, and they still require complicated postprocessing to integrate specific keypoint group, which is inconsistent with initial design ideal of anchor-free detectors. In addition, owing to the ambiguous appearances of tiny objects, it is difficult for the keypoint group to match corresponding point positions. Besides, corner or boundary keypoints are actually located in a weak semantic zone which locates among several coupled pixel-wise semantic relations. This phenomenon has a big risk to lead a poor pixel-wise interpretation of corners or boundaries and then makes the detector provide a bad detection performance. Next, except for keypoint group-based methods, the above-mentioned center point-based methods have proven that center point interpretation can be an effective way to boost optical remote sensing object detection in an anchor-free manner. Consequently, we concentrate on setting up a suitable semantic description to facilitate pixel-wise center point interpretation and achieve the detection of a multiscale object with irregular aspect ratios from optical remote sensing scenes.

B. Pixel-Wise Keypoint Interpretation

An effective semantic feature description is very important for anchor-free detectors, which can set up pixel-wise relations between each kind of object and BG. On one hand, the strong coupled pixel-wise relations can be set up to facilitate

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object center point interpretation and bboxes generation; on the other hand, the uncoupled pixel-wise relations can be used to avoid complex interference and false alarms. Thus, center point prediction can be improved by referring to the semantic segmentation technology [86]–[90]. Here, whether in object detection or semantic segmentation task, the FPN structure is widely employed as encoder-decoder architecture for deep and shallow layer feature fusion. Kong *et al.* [72] utilized FPN top-down single path way with lateral connections to construct semantic description and leverage center point interpretation in anchor-free detector of FoveaBox. Zhou *et al.* [69] adopted a deep layer aggregation (DLA) and hourglass structure to set up a semantic feature to achieve object center point prediction in CenterNet. Tian *et al.* [70] utilized the U-shape of top-down and bottom-up two pathways with cross-scale connections to generate a semantic description for centerness point prediction in FCOS. However, these ways only can generate naive a semantic description which is not friendly for modeling the pixel-wise relations from complex scenes. Thus, a lot of optimized ways [91]–[97] are designed and then applied for complex scenes. Yuan *et al.* [91] designed a multiresolution feature fusion to exploit the densely connected deconvolution layers with skip connections in vertical spatial sequence attention network (VSSA-Net), which can optimize the decoder to generate a more effective semantic description for small object detection. Li *et al.* [93] proposed a novel parallel down-up fusion network called PDF-Net for saliency object detection from optical remote sensing images, which takes full advantage of low- and high-level features and introduces the cross-path multiresolution to set up semantic description for multiscale saliency object detection. In addition, Li *et al.* [94] also proposed a nested network for saliency object detection in optical remote sensing scenes. The two-stream pyramid module containing L-shape and V-shape is introduced into encoder-decoder with nested connections, which can gradually fuse the multiscale details and semantic features for better saliency analysis of per pixels. As mentioned above, the more complex scenes request final semantic description to compound abundant receptive fields (RFs) from different layers to set up the semantic description. Especially for optical remote sensing scenes, how to construct an effective compound semantic is a worthy research topic for pixel-wise keypoint interpretation in anchor-free manners, which has to be continuously explored.

C. Multiscale Regression Layer With Hybrid Loss

In optical remote sensing object detection, there are many research works [47]–[52] utilizing pyramid structure as a multiscale regression layer to adapt vaster scale variances. Qiu *et al.* [48] designed a multiscale feature fusion gate module to construct four multiscale regression layers for optical remote sensing object detection, and then an aspect ratio attention module is constructed to adaptively select an appropriate aspect ratio of an object based on attention mechanism. Cheng *et al.* [51] proposed a cross-scale feature fusion module to generate five multiscale prediction layers which can adapt vaster scale variance of optical remote sensing objects. Wang *et al.* [52] designed a scale-invariant regression layer including three isolated scale branches, which divide full-scale information into fine-scale ranges to alleviate the great scale variance and achieve better object detection performance. In summary, the above-mentioned anchor-based detectors demonstrated that the multiscale prediction structure is very effective and has been widely applied to optical remote sensing object detection. Although several anchor-free-based detectors [81], [82] also adopt multiscale prediction structure as regression layer, they cannot generate accurate bboxes without any priori information (i.e., anchor scale and aspect ratio initialization). Thus, several works [81]–[83], [92] have considered utilizing a hybrid loss to promote a multiscale prediction layer to generate more accurate bboxes in an anchor-free manner, but what kinds of hybrid loss and their corresponding regression layer should be designed is an important problem to be explored.

III. Proposed Methodology

In this section, followed by the overall architecture of the proposed MSFC-Net shown in Fig. 1, first, a CSFF is introduced in Section III-A. Next, an FC regression layer and hybrid loss are introduced in Sections III-B and III-C.

A. Compound Semantic Feature Fusion

As mentioned before, to improve the ability of pixel-wise object center point interpretation from optical remote sensing scenes, an effective compound semantic description should be constructed to model the strong coupled or uncoupled semantic relations among each kind of pixel. In a general CNN network, each pixel on the feature layer can be seen as owning one kind of RFs. Then, as we know, a smaller RF is more effective for small-scale object identification, and a larger RF is very suitable for large-scale object classification. Thus, these different RFs' superpositions from different scale feature layers can generate a semantic description for each pixel interpretation, which has been demonstrated by the encoder-decoder structure in several semantic segmentation networks (e.g., U-Net [86], fully convolutional network (FCN) [87] pyramid scene parsing network (PSPnet) [88], global convolutional network (GCN) [89], and Deeplab-v3+ [90]).

As discussed in Sections I and II, optical remote sensing scenes usually include multiclass objects with vaster scale variances such as "building," "bridge," "roundabout," "car," "expressway," "airport" and so on, and the pixel-wise semantic relations among different category instances are generated by abundant RFs combination. For example, the large RFs can capture "building" "expressway" and "roundabout" information, and the above information can be regarded as strong-coupled semantic relations to support the pixel-wise interpretation of "car" whose information is captured by small RFs. Thus, for modeling compound semantic relations from optical remote sensing scenes, and inspired by EfficientDet [68] and neural architecture search (NAS)-FPN [97], which have proved cross-scale connections are very effective for multiscale feature fusion in FPN architecture, a CSFF architecture is proposed as shown in Fig. 1(a). Where, CSFF contains three parts: 1) BiFPN for multiscale semantic feature generation by using abundant RFs superposition;

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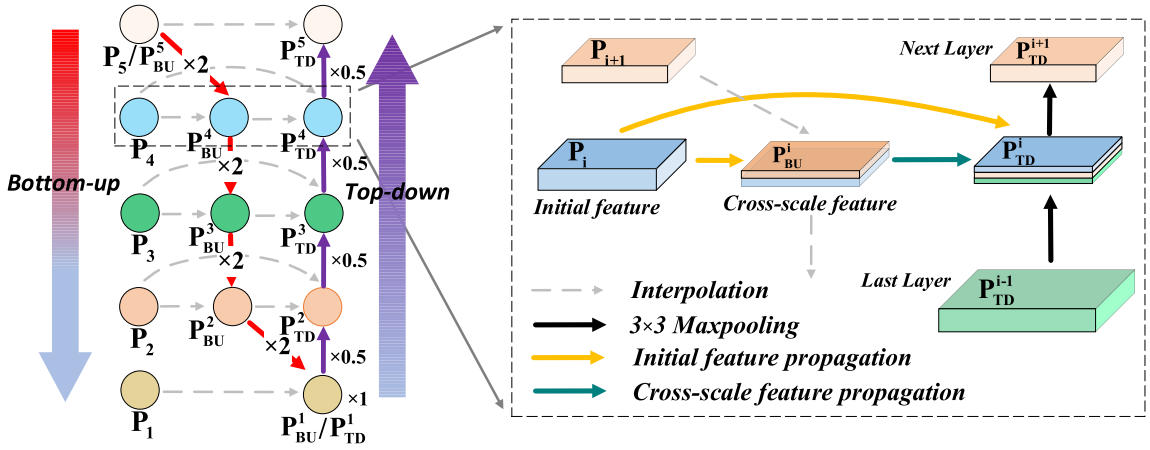


Fig. 2. Feature information propagation of the BiFPN structure. (a) Is the global two-path feature propagation way. (b) Is the local initial and cross-scale feature propagation method.

2) 3-D convolution for multiscale semantic feature integration; 3) compensatory fusion (CF) for final effective compound semantic construction. Next, as follows, three parts of CSFF architecture are elaborated.

1) BiFPN: Although the two-path feature propagation way of BiFPN as shown in Fig. 2(a) is similar to EfficientDet [68], the basic designing ideal is totally different. In EfficientDet [68], an effective multiscale feature fusion structure is designed to achieve abundant RFs combination from the backbone, and then, this way is integrated into FPN structure called BiFPN. Here, the method [68] repeated BiFPN three times to enable more high-level feature fusion, which is similar to the DLA structure in CenterNet [69]. However, their iterative feature propagation ways are too redundant and not friendly for DIOR. Despite, the iterative feature propagation way can suppress a large amount of interference produced by the uncoupled BG information, inevitably, it also would lose fine-scale details and discriminative object information at the last stage. Therefore, we rethink the iterative way of BiFPN in EfficientDet [68], and aim to set up the multiscale semantic features which are used for supporting the accurate center point interpretation. As a result, the two-path feature propagation way of BiFPN is utilized to achieve the RFs' combination and generate multiscale semantic descriptions among several adjacent scale layers. In Fig. 2(a), multiscale features $\mathbf{P_i}$ $i \in \{1, 2, 3, 4, 5\}$ from the backbone are fused by the two-path feature propagation way. Then, the bottom-up way is built to propagate large-scale global information from deep to shallow layer, and the top-down way is used to propagate the fine-scale detail information from shallow to deep layer. In detail, followed bottom-up way, a pair of adjacent lower and higher resolution features are fused by the interpolation and addition operations before performing a 1×1 convolution, and it can be expressed as follows:

$$\mathbf{P^i_{BU}} = \begin{cases} \mathbf{P_i}, & i = 5. \\ C\left(\mathbf{P_i} + U\left(\mathbf{P^{i+1}_{BU}}\right)\right), & i = 1, 2, 3, 4 \end{cases} \tag{1}$$

where, $C(\cdot)$ represents 1×1 convolution with stride 1, and $U(\cdot)$ denotes bilinear interpolation operation. i is the index of different scale layers in the two-path feature propagation way. $\mathbf{P_i}$ represents features from the backbone, and $\mathbf{P^i}_{\mathbf{BU}}$ indicates fused features in the bottom-up way. Next, followed top-down way, the higher resolution features, lower resolution features from bottom-up way and initial features from the backbone are fused by max-pooling and 1×1 convolution, which can be formulated as follows:

$$\mathbf{P^i_{TD}} = \begin{cases} \mathbf{P^i_{BU}}, & i = 1. \\ C(\mathrm{MP}(\mathbf{P^{i-1}_{TD}}) + \mathbf{P_i} + \mathbf{P^i_{BU}}), & i = 2, 3, 4. \\ C(\mathrm{MP}(\mathbf{P^{i-1}_{TD}}) + \mathbf{P_i}), & i = 5 \end{cases} \tag{2}$$

where, $\mathrm{MP}(\cdot)$ denotes 3×3 max-pooling, and $C(\cdot)$ represents 1×1 convolution with stride 1. Then, multiscale semantic features $\mathbf{P^i}_{\mathbf{TD}}$ in top-down way can be generated by the way of Fig. 2(b), because cross-scale and initial feature connections are introduced into two-path feature propagation way can facilitate abundant RFs combination.

2) 3-D Convolution: In common multiscale semantic feature fusion ways, the several multiscale semantic features would be rudely concatenated and compressed without considering their composability among different scales, and there is a semantic gap between every two scales of multiscale semantic features which are generated by BiFPN, especially for $\mathbf{P^5}_{\mathbf{TD}}$ and $\mathbf{P^1}_{\mathbf{TD}}$. Thus, in common feature fusion ways, complicated interscale interference is caused by directly fusing multiscale semantic features, which influences pixel-wise interpretation performance [100]. To avoid this semantic gap and construct a more effective compound semantic description, a 3-D convolution is designed for gradually fusing the multiscale semantic features. First, the multiscale semantic features $\mathbf{P^i}_{\mathbf{TD}}$, $i \in \{1, 2, 3, 4, 5\}$ from BiFPN are divided into three groups: $\{\mathbf{P^1}_{\mathbf{TD}}, \mathbf{P^2}_{\mathbf{TD}}\}$, $\{\mathbf{P^2}_{\mathbf{TD}}, \mathbf{P^3}_{\mathbf{TD}}, \mathbf{P^4}_{\mathbf{TD}}\}$, $\{\mathbf{P^4}_{\mathbf{TD}}, \mathbf{P^5}_{\mathbf{TD}}\}$ to make each group has a smaller semantic gap. Then, to eliminate smaller semantic gaps in each group, a 3-D convolution performs different types of convolutions on two or three adjacent multiscale semantic feature maps to achieve semantic feature alignment. Thus, as shown in Fig. 3, 3×3 convolutions with different strides and deformable convolution are employed to make up a 3-D convolution,

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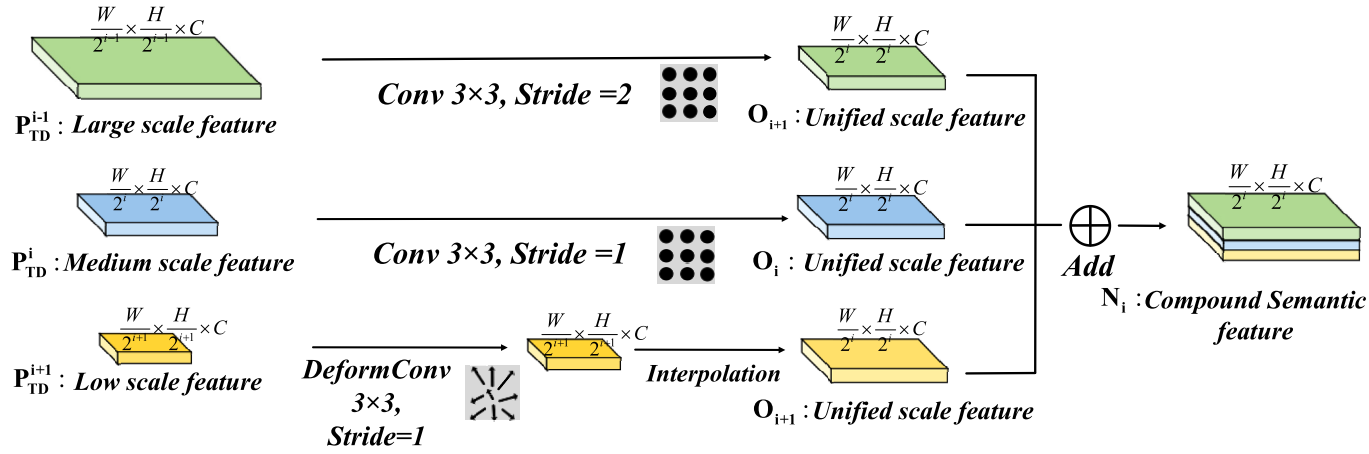


Fig. 3. Details of the 3-D convolution.

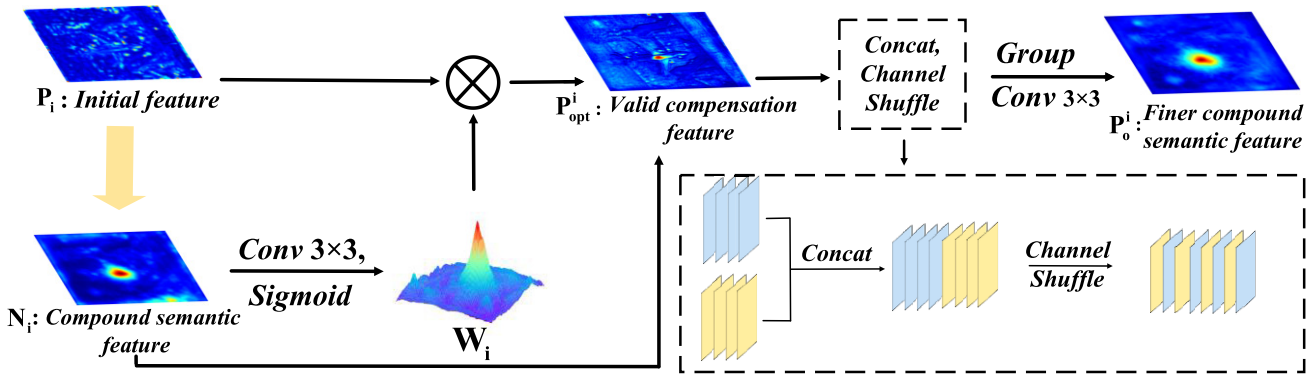


Fig. 4. Details of the CF operation.

which can be expressed as follows:

$$\mathbf{N_i} = \begin{cases} C_{S=2}\left(\mathbf{P_{TD}^{i-1}}\right) + C_{S=1}\left(\mathbf{P_{TD}^{i}}\right), & \text{if, } i-1=4. \\ C_{S=2}\left(\mathbf{P_{TD}^{i-1}}\right) + C_{S=1}\left(\mathbf{P_{TD}^{i}}\right) + U\left(\mathrm{DC}_{S=1}\left(\mathbf{P_{TD}^{i+1}}\right)\right) \\ \qquad \text{if, } i-1>1 \text{ and } i+1<5. \\ C_{S=1}\left(\mathbf{P_{TD}^{i}}\right) + U\left(\mathrm{DC}_{S=1}\left(\mathbf{P_{TD}^{i+1}}\right)\right), & \text{if, } i+1=2. \end{cases} \tag{3}$$

In (3), $\mathbf{N_i}$ $i \in \{1, 3, 5\}$ represents fused features, and i is the index that is used to indicate adjacent feature maps. $C_S(\cdot)$ denotes a 3 × 3 convolution with stride S. Next, $U(\cdot)$ is the bilinear interpolation operation, and $\mathrm{DC}_{S=1}(\cdot)$ means deformable convolution with stride 1. By using (3), we can see an example in Fig. 3, the features $\{\mathbf{P^{i-1}_{TD}}, \mathbf{P^{i}_{TD}}, \mathbf{P^{i+1}_{TD}}\}$ are regulated into the unified scale $\{\mathbf{O_{i-1}}, \mathbf{O_i}, \mathbf{O_{i+1}}\}$ by consistent sampling stride, and the fused $\mathbf{N_i}$ is obtained by adding two or three unified scale features. In general, the designed 3-D convolution can further enhance the semantic connectivity among adjacent layer features with smaller gaps and facilitate multiscale semantic feature fusion to model the coupled and uncoupled pixel-wise semantic relations among different scales.

3) Compensatory Fusion: After 3-D convolutions, the effective compound semantic descriptions can be constructed. However, through the complicated feature propagation of BiFPN and more reasonable multiscale semantic feature fusion of 3-D convolutions, there are a lot of discriminative details and tiny object information loss due to the multiple convolution information propagation ways. Thus, we consider compensating discriminative details and tiny object information by designing a CF operation. As shown in Fig. 1, we can see those three dashed curves from initial features of $\mathbf{P_1}$, $\mathbf{P_3}$, and $\mathbf{P_5}$ are introduced into three effective compound semantic features of $\mathbf{N_1}$, $\mathbf{N_3}$, and $\mathbf{N_5}$. Then, the designed CF operation can be illustrated in Fig. 4 and formulated by the following equations:

$$\mathbf{W_i} = \mathrm{SIG}(C_{S=1}(\mathbf{N_i})) \tag{4}$$

$$\mathbf{P^i_{opt}} = \mathbf{W_i} \times \mathbf{P_i} \tag{5}$$

$$\mathbf{P^i_o} = \mathrm{ReLU}\left(C_{s=1}\left(\mathrm{CS}\left(\mathrm{CONCAT}\left(\mathbf{N_i}, \mathbf{P^i_{opt}}\right)\right)\right)\right). \tag{6}$$

In (4), $\mathrm{SIG}(\cdot)$ denotes the sigmoid function, and $C_{S=1}(\cdot)$ denotes a 3 × 3 convolution with stride 1. Then, they are applied on compound semantic features $\mathbf{N_i}$ to get activated coefficients of coupled and uncoupled semantic relations $\mathbf{W_i}$. Then, in Fig. 4, the activated coefficients of $\mathbf{W_i}$ are multiplied by initial features $\mathbf{P_i}$ from the backbone to obtain $\mathbf{P^i_{opt}}$ which is used to select valid details and tiny object information to be the compensating of $\mathbf{N_i}$. Next, followed a concatenation operation $\mathrm{CONCAT}(\cdot)$ and a channel shuffling $\mathrm{CS}(\cdot)$ in (6), valid compensation information $\mathbf{P^i_{opt}}$ fully interacts with $\mathbf{N_i}$. Then, the 3 × 3 group convolution with stride 1 and rectified linear unit (ReLU) activation function are applied to obtain the final effective compound semantic features $\mathbf{P^i_o}$. Finally, by utilizing the designed CF operation, three effective compound semantic features can be further improved, and it not only adaptively select valid details or tiny object information

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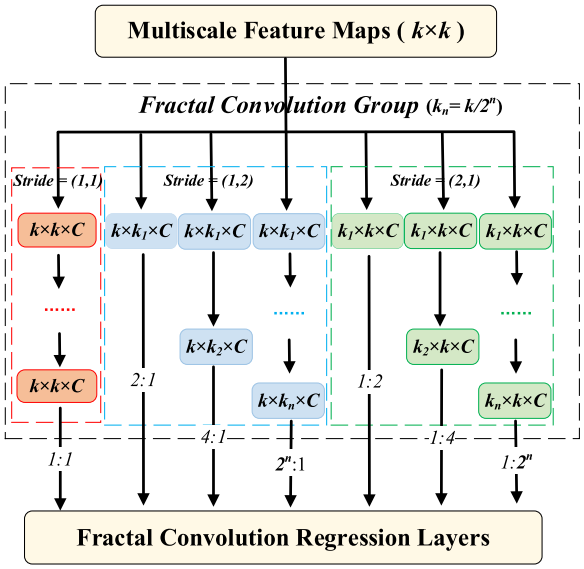


Fig. 5. Construction principle of the FC regression layers.

for pixel-wise object center point interpretation, but also can avoid false alarms of uncoupled semantic relations from BG. Besides, the three constructed effective compound semantic features have strong pixel-wise interpretation ability, and they can be easily extended into the semantic segmentation task for land use classification.

B. FC Regression Layers

As shown in Fig. 1(b), followed by the CSFF, three generated compound semantic features are utilized to individually set up 128 × 128 × 128 and 256 × 256 × 128 feature maps, which are prepared for pixel-wise object center point prediction and bboxes regression. However, since there is no priori information (e.g., scales and aspect ratios) supporting for accurate bboxes regression, it would make the detector hardly adapt the multiscale bboxes regression with irregular aspect ratios. To address this issue, we refer to a fractal system [101], [102] and design an FC regression layer to adaptively fit scale and aspect ratio changes. Besides, under extreme aspect ratio cases, it also can facilitate center point prediction and bboxes regression by ensuring the same feature sampling rates along vertical and horizontal directions in an anchor-free manner.

Conceptually, a fractal is an infinitely complex pattern, which can be seen as a multiscale system with self-similarity characteristics across different scales. Then, multiscale traits are presented by object self-scale changes and multiplatform observations from optical remote sensing imagery, and self-similarity is reflected by the similar variance of irregular aspect ratios in local and global scale ranges. Thus, fractals can be used to model the regression layer of the detector for center point prediction and multiscale bboxes regression with irregular aspect ratios. In Fig. 5, inputs of FC regression layer are multiscale feature maps, and FC group is different from conventional convolution with a consistent stride, as these use different strides of {(1,1), (1,2), (2,1)} in terms of the vertical and horizontal dimensions to perform convolution operations repeatedly. Then, an FC regression layer would contain several

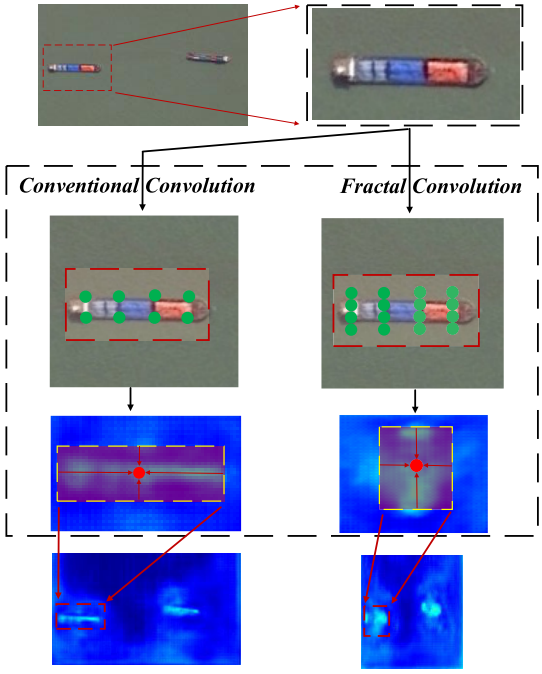


Fig. 6. Different in the high aspect ratio object feature collections of conventional convolutions and FCs.

irregular aspect ratio feature maps that are generated by the FC group; obviously, an FC regression layer is a never-ending pattern. Thus, to tradeoff the computing resources and efficiency and then considering medium- and large-scale objects have the more obvious irregular aspect ratios, the 128 × 128 × 128 feature map is used to generate the irregular and unequal size regression feature maps. In addition, 256 × 256 × 128 feature map is used for bboxes regression of tiny object which has a less aspect ratio variance and the requirement of fine-grained center point prediction. Consequently, the high spatial resolution feature map of 256 × 256 × 128 combined with several irregular and unequal size feature maps can be used to build FC regression layers for adaptively fitting multiscale bboxes regression with irregular aspect ratios. Then, for extreme aspect ratio cases [i.e., ships, bridges, and large vehicles (LV)], an example is given in Fig. 6, where we can see that along with vertical and horizontal directions, the ship with a high aspect ratio has a sparser feature collection at narrow height when the conventional convolution is used, so it would obtain insufficient feature information along vertical direction for accurate center point prediction and bboxes regression. Next, from Fig. 6, the FC group can ensure a consistent sampling rate for feature collection to some extent under extreme aspect ratio conditions. Therefore, multiclass objects with vaster scale variances and irregular aspect ratios can be automatically assigned to a suitable feature map from the set up FC regression layers, and then accurate center point prediction results and its corresponding bboxes regression can be obtained reasonably. Next, an FC regression layer also can facilitate the rapid convergence of the proposed MSFC-Net, whose principle can be expressed as follows:

$$\text{loss}_w = \left|w - w'\right| = \left|w - s_i \times w_i\right| \tag{7}$$

$$\text{loss}_h = \left|h - h'\right| = \left|h - s_i \times h_i\right|. \tag{8}$$

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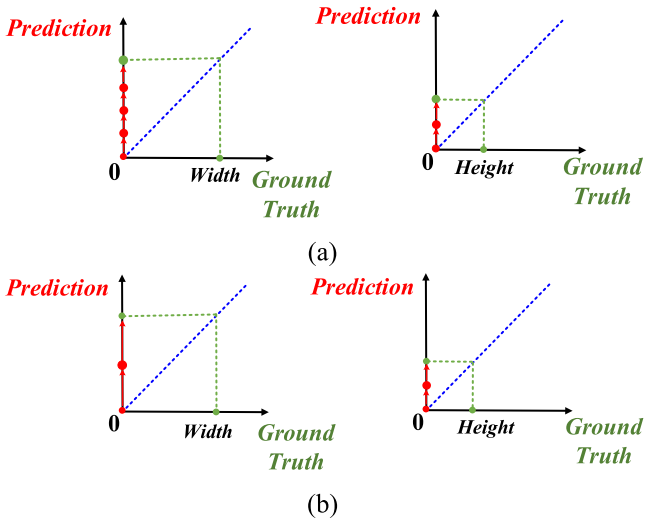


Fig. 7. Regression difference analysis between the conventional convolution and FC. (a) Represents the regression situation with conventional convolution (i.e., stride_w:stride_h = 1:1). (b) Represents the regression situation with FC (i.e., stride_w:stride_h = 2:1).

In (7) and (8), loss_w and loss_h represent the location losses in the regression process. w and h indicate the width and height of GTs, and w_i and h_i represent the predicted width and height, respectively. Here, w' and h' indicate that w_i and h_i are remapped to the original image by a scale factor s_i, and i is an index for the set up FC regression layer. Then, if it appears extreme aspect ratio case of $w \gg h$, as in the conventional convolution with the same stride, the convergence speeds of loss_w and loss_h are very different, which can be illustrated as Fig. 7(a). This situation affects the efficiency of model learning and makes it difficult for the model to converge. In contrast, an FC group with different strides for the width and height regressions is shown in Fig. 7(b), it can ensure detector converge rapidly and synchronously.

C. Hybrid Loss for Model Training

To further facilitate the proposed MSFC-Net training process, a specific hybrid loss designed for the FC regression layer is defined as follows:

$$L = \lambda_C L_C + \lambda_{\text{offset}} L_{\text{offset}} + \lambda_{\text{size}} L_{\text{size}}. \tag{9}$$

Equation (9) contains the hybrid loss function L including the center point loss L_c, size loss L_{size}, and offset loss L_{offset}, and the weight parameters λ_c, λ_{size}, and λ_{offset} are equal to 1, 0.1, and 1, respectively. First, the center point classification loss L_c is designed in the same way as in CenterNet [69], and it can be expressed as follows:

$$L_c = \frac{-1}{N} \begin{cases} \left(1 - \hat{Y}_{ij_c}\right)^{\alpha} \log(\hat{Y}_{ij_c}), & \text{if } Y_{ij_c} = 1 \\ \left(1 - Y_{ij_c}\right)^{\beta} \left(\hat{Y}_{ij_c}\right)^{\alpha} \log(1 - \hat{Y}_{ij_c}), & \text{otherwise.} \end{cases} \tag{10}$$

In (10), $\hat{Y}_{ij_c} \in [0, 1]^{(W/R_W)\times(H/R_H)\times C}$ is the classification score for class c in predicted heatmap location (i, j), and R_W and R_H can help remap the predicted coordinates of the heatmaps to the original input represent the width and height of original input images. W and H represent the width and height of original input images. In the set up FC regression layer, when the width and height of a given heatmap are equal, $R_W = R_H$; otherwise, $R_W \neq R_H$. $Y_{ij_c} \in [0, 1]^{(W/R_W)\times(H/R_H)\times C}$ indicates GTs; when $Y_{ijc} = 1$, the center points belong to positive samples, and when $0 \leq Y_{ijc} < 1$, the center points belong to negative samples. Next, to select appropriate center point samples for model training, Y_{ijc} in (10) can be expressed as follows:

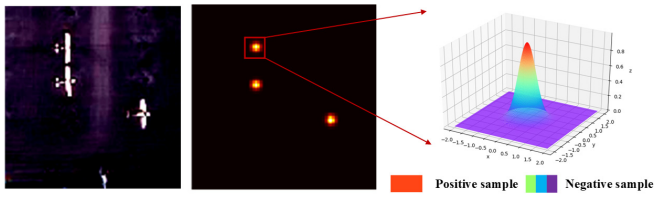


Fig. 8. Definitions of positive and negative training samples.

$$Y_{ij_c} = \exp\left(-\frac{(x - \tilde{p}_i)^2 + \left(y - \tilde{p}_j\right)^2}{2\sigma_p^2}\right). \tag{11}$$

In (11), x and y are neighboring coordinates compared with the positive center coordinates $(\tilde{p}_i, \tilde{p}_j)$, where $\tilde{p}_i = \lfloor (p_i/R_W) \rfloor$ and $\tilde{p}_j = \lfloor (p_j/R_H) \rfloor$. (p_i, p_j) represents the center point of the GT in the original image. Next, $\sigma_p = (2 \times \max(w, h) + 1)/6$ defines the size-adaptive object standard deviation of the Gaussian kernel Y_{ijc}, and w and h are the width and height of the corresponding GTs, respectively. Thus, this Gaussian kernel can be used to balance the positive and negative training samples, as illustrated in Fig. 8.

Second, since the output stride of downsampling would result in discretization error, the center point offset is predicted additionally. In detail, the second term offset loss in (9) can be expressed as follows:

$$L_{\text{offset}} = \frac{1}{N} \sum_{p} \left| \hat{O} - P_{\tilde{p}} \right|. \tag{12}$$

As shown in (12), this is the L_1 loss, and N is the number of center points in the image. $\hat{O} \in R^{(W/R_W)\times(H/R_H)\times 2}$ represents the predicted offsets, and $P_{\tilde{p}}$ represents the actual offsets $((p_i/R_W) - \tilde{p}_i, (p_j/R_H) - \tilde{p}_j)$. In general, combining the L_c and L_{offset} constraint terms in (9) can facilitate more accurate center point predictions.

Third, due to without any priori information of anchor scales and aspect ratios, the proposed MSFC-Net directly regresses and predicts bboxes based on center point prediction, which has large differences in the diversity of object scales. Large scale objects provide more contributions to the width and height regressions in the training phase, and the widths and heights of tiny objects would take a big risk of being ignored. Therefore, for achieving scale-invariant regression, GIoU [107] loss is introduced into the third term L_{size} in (9). Here, the GIoU [107] loss can make the width and height regressions have equivalent contributions under different scales because they have the same sensitivity to the IoU constraints. Moreover, although the GIoU [107] can perceive scale variances whether encountering a tiny, medium, or large object, it cannot constrain the proposed MSFC-Net to generate accurate bboxes for detection. Therefore, the L_1 loss of the width and height is also needed in L_{size}. Based on the above

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considerations, L_{size} can be defined as follows:

$$\begin{aligned} L_{\text{size}} &= \frac{1}{\lambda_{\text{size}}} \cdot L_{\text{GIoU}} + L_{L_1} \\ &= \frac{1}{N} \sum_{k=1}^{N} \left\{ \frac{1}{\lambda_{\text{size}}} \cdot \left[1 - \left(\text{IoU}_k - \frac{|A_k - U_k|}{A_k} \right) \right] + \left| \hat{S}_k - S_k \right| \right\}. \end{aligned} \tag{13}$$

In (13), IoU_k is an intersection over the union ratio between the predicted and GT bboxes, and A_k is the minimum closure area of the predicted and GT bboxes. U_k is the union area of the predicted and GT bboxes. Then, in (13), the second term is the L_1 loss, which measures the loss produced by the predicted values $\widehat{S}_k = (\widehat{h}, \widehat{w})$ and GTs' $S_k = (h, w)$, where w and h represent the GT's width and height, and $\hat{h}$ and $\hat{w}$ indicate the predicted width and height. In general, the comprehensive hybrid loss of L_c, L_{offset}, and L_{size} in (9) can make the MSFC-Net converge rapidly and reasonably in the training phase.

IV. Experiments and Analysis

In this section, the experimental details are elaborated; these include datasets and implementation details, evaluation metrics, a comparison analysis, and ablation experiments.

A. Datasets and Implementation Details

In our experiments, we evaluate comparison methods on four optical remote sensing datasets to demonstrate the effectiveness of the proposed MSFC-Net. We employ the DOTA [98] and DIOR [99] datasets as object detection benchmarks and utilize the International Society for Photogrammetry and Remote Sensing (ISPRS) [103] and gaofen image dataset (GID) [104] land use classification datasets to demonstrate that the designed CSFF can provide the most effective compound semantic description for pixel-level object center point prediction.

1) DOTA Dataset: DOTA [98] is a large-scale object detection dataset that includes 2806 aerial images with different sizes ranging from 800 × 800 to 4000 × 4000 pixels. It contains 188 282 instances with different scales and aspect ratios. There are 15 object categories: plane (P), baseball diamond (BD), bridge (B), ground track field (GTF), small vehicle (SV), LV, ship (S), tennis court (TC), basketball court (BC), storage tank (ST), soccer ball field (SBF), roundabout (R), harbor (HB), swimming pool (SP), and helicopter (H). The dataset is divided into three subsets: training, validation, and testing. In our experiments, the proposed MSFC-Net is evaluated on the horizontal bounding box (HBB) task, and we resize the training set into multiple scales, ×1.2, ×1, and ×0.4, and crop them into 512 × 512 subimages, where each 512 × 512 training image has a 200-pixel overlap. Finally, following this rule, we can obtain 54 500 images for training and use validation as the comparison standard.

2) DIOR Dataset: To further verify the effectiveness of the proposed MSFC-Net for object detection in large-scale data and multiple categories, we additionally introduce DIOR [99], which is currently the largest dataset for optical remote sensing object detection. It contains 23 463 images with 192 472 instances involving 20 object categories, such as airplane (AL), airport (AT), baseball field (BF), BC, bridge (B), chimney (C), dam (D), expressway service area (ESA), expressway toll station (ETS), golf course (GC), GTF, HB, overpass (O), ship (S), stadium (SD), ST, TC, train station (TS), vehicle (V), and windmill(W). In addition, the size of images in DIOR [99] is 800 × 800 pixels, and the spatial resolutions range from 0.5 to 30 m. Then, 11 725 images are randomly selected from the dataset as training and validation sets. The remaining 11 738 images are employed for testing. Last, we resize the training set into multiple scales, ×1.28, ×1, and ×0.64, and crop them into 512 × 512 subimages. Through the above operations, we obtain 38 109 images for training.

3) ISPRS Dataset: To demonstrate the effectiveness of the proposed CSFF for constructing a compound semantic feature to achieve accurate pixel-level prediction, the ISPRS Vaihingen 2-D dataset [103] is adopted as the semantic segmentation evaluation benchmark. ISPRS consists of airborne images of Vaihingen. It contains 33 tiles of different sizes, 16 of which have publicly available GTs, and these pixels in the tiles are labeled with the following six classes: Building (B), Low vegetation (LV), Tree (T), Car (C), impervious surfaces (IS), and BG. To unify the input scale, we crop tiles into 512 × 512 pixels and choose 3576 images for training and 7438 images for testing.

4) GID Dataset: To further demonstrate that the proposed CSFF architecture can provide an effective compound semantic feature to set up pixel-wise strongly coupled or uncoupled semantic relations in complex optical remote sensing scenes, GID [104], named the "Gaofen Image dataset," is also employed as a semantic segmentation benchmark, and it covers areas of approximately 50 000 km^2 in China and includes six categories: built-up (BU), farmland (FL), forest (F), meadow (M), BG, and water (W) with complex remote sensing scenes. For the GID [104] dataset, we resize the 3750 images to 512 × 512 pixels and split them into 2000 training and 1750 validation images.

5) Implementation Details: The backbone of the proposed MSFC-Net can be replaced by any widely used classification network, and then we utilized the ResNeSt-101 which is pretrained on ImageNet [56] to get the state-of-the-art performance. In addition, the inputs of all comparison detectors are regulated into 512 × 512 pixels, and the Adam optimizer [105] is adopted in the training phase. Whether for object detection or land use classification tasks, random scaling, cropping, and flipping data augmentation operations are employed. Except for the above-mentioned data argumentation methods, the mosaic enhancement is adopted for Yolo-v4 [63] and v5 [64]. Finally, in the object detection task, the training batch size is set to ten and the initial learning rate is set to 6.25e-5, and it decreases by 0.5 at 50, 90 epochs; in the land use classification task, the batch size is also set to 16 and the learning rate is initialized as 1e-3 and decreases by 0.5 per 70 epochs. Then, a multiscale testing strategy is adopted in all comparison detectors, and all models are implemented by PyTorch 1.1.0 and executed on a TITAN RTX GPU.

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TABLE I
OBJECT DETECTION PERFORMANCE COMPARISONS ON DOTA

Method	*mAP*	*mF1*	*P*	*BD*	*B*	*GTF*	*SV*	*LV*	*S*	*TC*	*BC*	*ST*	*SBF*	*R*	*HB*	*SP*	*H*
FasterRCNN [59]	56.13	64.82	76.15	63.70	29.60	54.86	28.59	62.59	67.70	86.89	50.10	67.33	55.84	40.91	65.71	43.64	48.37
Yolo-v3 [62]	65.67	69.34	93.91	67.78	45.93	51.92	50.12	60.67	85.56	93.88	66.82	83.47	52.45	45.01	74.03	55.85	56.68
Yolo-v4 [63]	73.94	73.58	97.31	74.98	49.58	46.40	**83.60**	81.44	93.49	93.74	70.66	89.30	50.61	73.94	80.86	75.83	47.42
Yolo-v5 [64]	74.22	74.67	96.99	71.18	45.01	59.28	78.93	79.92	92.29	93.33	60.87	85.80	59.36	75.40	76.65	80.25	57.99
RetinaNet [66]	63.27	66.17	72.99	68.17	**65.96**	**76.22**	25.51	62.78	68.59	84.20	62.33	51.31	57.78	**80.87**	65.96	57.81	48.50
YOLT [25]	61.44	67.85	87.20	60.50	22.09	37.54	70.75	70.48	84.43	93.94	70.93	78.96	52.70	57.62	56.58	55.76	22.18
RFB-Net [67]	64.03	66.04	87.75	49.76	35.74	36.35	78.00	77.53	90.05	93.84	63.83	89.54	33.57	67.93	63.21	72.78	20.53
EfficientDet [68]	61.96	69.01	93.10	66.32	44.22	51.14	38.87	52.39	81.57	92.18	64.63	78.27	52.12	42.74	72.25	44.54	55.05
LV-Net [94]	67.80	69.21	96.79	62.98	40.61	53.93	72.02	77.83	89.75	95.09	64.41	87.07	56.31	57.43	63.42	67.64	31.68
FBR-Net [81]	73.24	72.77	95.76	67.52	41.03	46.17	71.34	72.76	88.30	95.42	68.59	86.54	73.35	69.22	80.16	75.46	66.98
Centernet-dla [69]	70.68	71.46	95.72	67.23	42.07	64.46	71.08	81.45	91.54	94.94	59.63	85.05	48.05	68.09	75.29	74.04	41.56
Centernet [69]	73.94	74.02	97.37	78.56	45.39	53.39	62.16	80.24	90.30	94.58	66.11	85.75	64.86	69.02	78.86	75.63	66.82
FCOS [70]	76.95	76.83	97.09	77.46	48.78	55.14	74.28	81.69	91.92	**96.02**	74.18	88.70	**75.82**	71.31	**85.60**	**76.74**	59.48
FoveaBox[72]	71.81	72.46	95.38	69.04	40.76	58.49	72.27	70.15	89.94	94.90	63.92	88.39	50.02	71.19	79.91	59.60	59.29
CornerNet [71]	63.33	66.77	93.58	61.10	24.48	63.02	58.31	71.89	64.13	92.39	57.87	81.89	50.86	53.94	74.82	52.87	48.78
DPAFPN [80]	69.00	70.17	93.99	66.32	44.68	51.87	71.23	78.12	89.62	95.21	71.57	87.03	24.15	70.33	65.98	74.37	50.44
MSFC-Net (our)	**80.26**	**79.33**	**98.00**	**79.67**	55.76	75.48	76.13	**83.40**	**94.14**	95.85	**74.27**	**90.65**	74.89	74.36	85.02	74.21	**71.79**

B. Evaluation Metrics

To quantify the performances of different detectors and land use classification semantic segmentation networks, several evaluation metrics are used such as precision, recall, average precision (AP), mAP mF1-score (mF1), pixel accuracy (PA), and mean IoU (mIoU). The relevant formulas are listed below.

1) Object Detection:

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{14}$$

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{15}$$

$$\text{mF1} = \frac{1}{C}\sum_{i=1}^{C}\frac{2 \times \text{recall} \times \text{precision}}{\text{recall} + \text{precision}} \tag{16}$$

$$\text{mAP} = \frac{1}{C}\sum_{i=1}^{C}\text{AP}_i. \tag{17}$$

In (14) and (15), recall and precision are defined by TP, FP, TN, and FN, which represent true positives, false positives, true negatives, and false negatives, respectively. mF1 is calculated by recall and precision, as illustrated in (16), and related to (17), AP and mAP refer to PASCAL visual object classes (VOC) criteria [106]. AP means the integral area under the precision and recall (P–R) curves, and mAP indicates the average AP of C categories.

2) Land Use Classification:

$$\text{PA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{18}$$

$$\text{mIoU} = \frac{1}{C}\sum_{i=1}^{C}\frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}}. \tag{19}$$

In (18) and (19), PA and mIoU are used to evaluate the land use classification task, and TP, FP, TN, and FN denote the true positives, false positives, true negatives, and false negatives of pixel-level classification results, respectively.

C. Comparison Analysis

To verify that the proposed MSFC-Net is superior to the other state-of-the-art methods, several famous CNN-based detectors are employed as comparisons including eight two- or one-stage anchor-based detectors (e.g., Faster-RCNN [59], Yolo-v3 [62], Yolo-v4 [63], Yolo-v5 [64], RetinaNet [66], YOLT [25], RFB-Net [67], and EfficientDet [68]) and eight anchor-free detectors (e.g., two versions of CenterNet -DLA and -Res [69], FCOS [70], CornerNet [71], LV-Net [94], feature balancing and refinement network (FBR-Net) [81], FoveaBox [72], and dense path aggregation feature pyramid network (DPAFPN) [80]). Several comparison experiments are carried out on the DOTA [98] and DIOR [99], and these results are reported in Tables I and II. According to Tables I and II, we can see that the proposed MSFC-Net can provide superior performance, with 80.26% mAP and 79.33% mF1 on DOTA [98] and 70.08% mAP and 73.45% mF1 on DIOR [99], thereby outperforming the state-of-the-art methods. Next, we analyze these comparison results on DOTA [98] and DIOR [99], individually.

1) DOTA: As reported in Table I, we can see that from lines 1 to 8, there are eight anchor-based detectors being selected as comparisons and tested on DOTA [98], and the other eight anchor-free detectors are also tested and presented in lines 9 to 16. In general, the anchor-free detectors can perform better detection results than anchor-based detectors' on DOTA [98] because the anchor-based detectors are restricted by the fixed anchor region classification and localization detection mechanism. Besides, abundant and continuous scale variances in a large scope would make the discrete predefined anchors hard to cover full-scale object information based on the single-scale prediction layer. Here, from line 1 in Table I, although multiscale anchors are predefined for Faster-RCNN [59], it only utilizes the last layer of backbone for region proposal network (RPN) which cannot adapt the multiscale object feature description and prediction.

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TABLE II
OBJECT DETECTION PERFORMANCE COMPARISONS ON DIOR

Method	*mAP*	*mF1*	*AL*	*AT*	*BF*	*BC*	*B*	*C*	*D*	*ESA*	*ETS*	*GC*	*GTF*	*HB*	*O*	*S*	*SD*	*ST*	*TC*	*TS*	*V*	*WM*
FasterRCNN [59]	53.61	64.41	51.35	61.62	62.21	80.66	26.96	74.18	37.26	53.46	45.12	69.63	61.81	43.76	48.97	56.14	41.82	39.56	73.88	44.74	33.98	65.32
Yolo-v3 [62]	60.60	66.51	68.86	55.39	66.74	87.14	35.01	73.96	34.63	56.15	49.81	67.98	69.59	55.16	52.51	87.71	42.05	68.93	84.56	33.62	49.82	72.37
Yolo-v4 [63]	66.39	71.28	79.25	71.23	70.23	88.66	39.20	77.08	48.64	58.24	58.34	75.45	71.23	53.84	57.74	**90.12**	51.04	77.54	83.34	46.66	55.13	75.05
Yolo-v5 [64]	68.57	**75.89**	87.28	61.67	73.75	90.04	42.64	77.54	55.20	**63.81**	63.22	66.89	**78.03**	**58.17**	**58.12**	87.78	54.32	79.32	89.71	50.24	53.96	79.63
RetinaNet [66]	65.22	71.20	63.46	47.84	**83.31**	78.36	**48.19**	67.93	49.42	53.17	73.60	66.25	59.06	47.55	45.67	72.03	**82.38**	**80.68**	**90.13**	54.98	47.71	**92.92**
YOLT [25]	60.29	68.51	64.77	68.98	62.85	87.89	32.37	71.57	45.86	54.93	55.86	65.68	66.35	49.93	49.97	87.74	30.36	73.39	82.06	29.95	52.45	73.96
RFB-Net [67]	62.59	67.35	**93.88**	48.36	80.27	77.78	40.78	67.56	47.43	48.54	**66.96**	37.37	54.96	48.24	43.17	66.18	77.95	79.97	89.17	**55.64**	40.46	88.37
EfficientDet [68]	62.22	71.66	72.47	68.37	64.60	87.01	33.55	74.54	43.69	60.16	55.43	72.60	66.90	46.96	52.98	86.29	37.61	70.85	81.20	43.42	50.27	75.45
LV-Net [94]	45.89	53.46	67.46	4.84	66.10	86.23	18.30	67.10	8.58	24.86	38.65	29.32	50.12	14.30	33.65	84.43	32.94	73.72	81.78	13.30	51.08	71.02
FBR-Net [81]	60.01	67.05	68.74	56.42	67.86	85.82	31.20	78.17	46.46	49.21	51.62	57.31	67.74	41.81	49.92	75.87	60.93	66.86	82.17	45.04	42.72	73.95
Centernet-dla [69]	63.18	67.38	78.61	56.50	76.11	88.15	33.24	77.19	41.05	47.37	55.53	71.38	72.56	23.02	52.75	89.84	54.00	78.62	86.23	46.13	**57.86**	77.45
Centernet [69]	63.86	69.53	73.58	57.98	69.73	88.46	36.20	76.88	47.90	52.66	53.90	60.54	62.62	45.68	52.60	88.21	63.74	76.21	83.66	51.32	54.43	79.53
FCOS [70]	61.17	65.95	73.50	68.01	69.86	85.11	34.66	73.60	49.33	52.06	47.56	67.21	68.67	46.31	51.06	72.24	59.84	64.61	81.17	42.72	42.17	74.78
FoveaBox [72]	60.51	67.09	68.93	42.80	68.27	86.58	37.46	79.11	34.97	43.94	52.98	53.52	66.23	44.39	46.03	87.59	72.68	74.55	85.09	40.95	47.87	76.19
CornerNet [71]	56.66	64.02	67.36	65.54	79.14	81.44	30.97	66.65	43.96	59.55	49.16	56.44	67.36	28.54	45.93	56.54	60.93	46.24	84.52	45.54	35.14	61.42
DPAFPN [80]	61.77	66.54	74.77	40.32	68.36	87.82	39.08	**79.44**	44.03	48.84	57.34	66.73	68.97	35.22	54.53	86.73	56.92	72.03	83.33	41.53	49.52	79.81
MSFC-Net (our)	**70.08**	73.45	85.84	**76.24**	74.38	**90.10**	44.15	78.12	**55.51**	60.92	59.53	**76.92**	73.68	49.55	57.24	89.62	69.21	76.52	86.74	51.82	55.23	84.31

TABLE III
ABLATION EXPERIMENTS OF THE PROPOSED MSFC-NET

Method	*CSFF Module*			*Hybrid loss*	*FC Regression Layers (1:2)*	*FC Regression Layers (1:4)*	*mAP(%)*
	BiFPN	*3-D Conv*	*CF Operation*				
Baseline	✓						74.95
	✓	✓					75.92
	✓		✓				77.21
	✓	✓	✓				78.12
	✓	✓	✓	✓			78.49
	✓	✓	✓	✓	✓		79.79
	✓	✓	✓	✓	✓	✓	80.13
MSFC-Net	✓	✓	✓	✓		✓	**80.26**

Especially for usually occurred tiny object detection situation, a large RF of the last feature layer in the backbone would provide the excessive context information, but it cannot support the class and location prediction for a tiny object in a fixed anchor region, thus Faster-RCNN [59] only can provide 56.13% mAP, 64.82% mF1. Then, the same problem also occurs in other anchor-based detectors. For example, despite multiscale regression layer and multiscale anchor initialization are utilized in Yolov3 [62], RetinaNet [66], YOLT [25], and RFB-Net [67], they also cannot provide satisfactory results because these anchor-based methods are hard to deal with the anchor initialization for vast scale changes and the missing detection situations in densely arranged object detection caused by NMS. Then, we can see that Yolo v4 [63] and v5 [64] can perform the better detection results on DOTA [98], which can obtain 73.94% mAP, 73.58% mF1 and 74.22% mAP, 74.67% mF1, individually. Where, they benefit from the data augmentation strategy of mosaic, which can enrich the diversity of training data (i.e., this allows the anchor-based detectors to detect the object outside its normal context) and offset the imbalance samples of different scales and aspect ratio variances in training data. Next, a powerful data augmentation strategy combined with K-means anchor clustering makes Yolo-v5 [64] obtain better performance than other anchor-based detectors. Then, except for the anchor-based detectors, we can see that in Table I, the center point interpretation-based anchor-free detectors of FBR-Net [81], CenterNet-DLA [69], CenterNet [69], FCOS [70], and FoveaBox [72] also can provide the better performance without a specific data augmentation strategy. Nevertheless, these anchor-based and anchor-free detectors both have the lower mAP and mF1 than the proposed MSFC-Net, because an effective compound semantic guided keypoint interpretation mechanism and fractal system regression layer can boost the detector to get superior performance than state-of-the-art methods.

2) DIOR: As illustrated in Table II, the more challenging optical remote sensing object detection dataset DIOR [99] is utilized to test the proposed MSFC-Net and other comparisons. As described before, the DIOR [99] dataset has more instances with complex interference than DOTA [98] datasets, thus for 20 instances in DIOR [99], it has to set up more complex and effective coupled semantic relations for pixel-wise center point interpretation. From lines 9 to 16 in Table II, these anchor-free detectors have the descending of mAP and mF1 than DOTA [98] dataset because of their unsuitable semantic relation modeling. Although these detectors of

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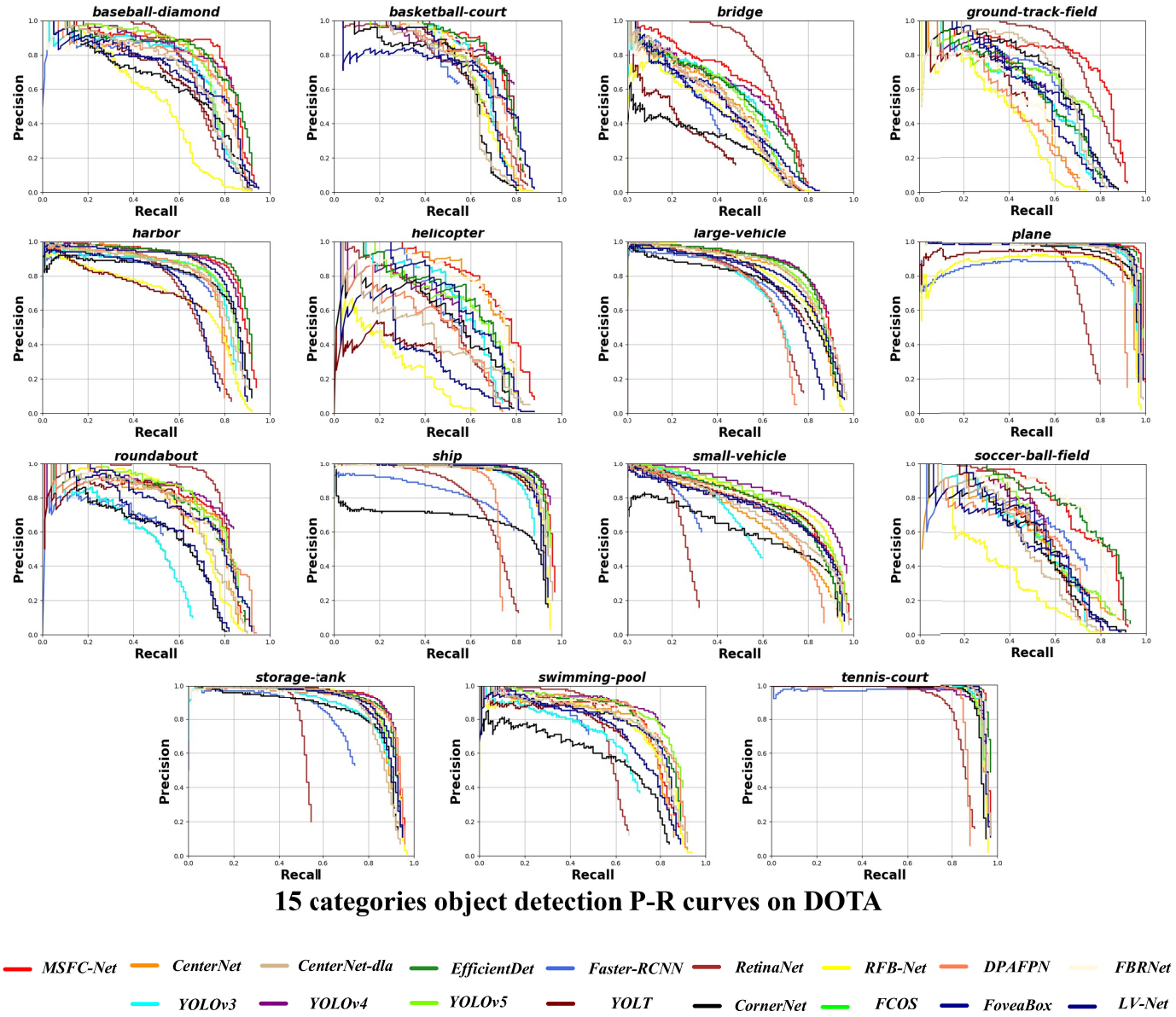


Fig. 9. *P*–*R* curves of the comparisons on DOTA [98].

TABLE IV
SEMANTIC SEGMENTATION PERFORMANCE COMPARISONS ON ISPRS

Method	*BG* (%)	*IS* (%)	*T* (%)	*B* (%)	*C* (%)	*LV* (%)	*PA* (%)	*mIoU* (%)
Deeplabv3+ [90]	88.7	37.1	87.7	**93.0**	81.5	**76.8**	86.5	67.4
GCN [89]	89.1	36.7	88.4	92.2	72.3	75.9	86.3	65.3
FCN [87]	86.9	16.6	87.6	89.2	57.6	73.7	83.8	59.0
CSFF (our)	**91.2**	**49.4**	**88.5**	92.8	**81.8**	75.5	**87.3**	**70.9**

LV-Net [94], FBR-Net [81], and DPAFPN [80] are considered different detection characteristics of optical remote sensing, they did not pay attention to model the more complex coupled or uncoupled semantic relations from optical remote sensing scenes. Thus, comparing with our proposed MSFC-Net, the results in Table II line 9, 10, and 16 demonstrate that the general encoder-decoder way (e.g., the two paths of U-shape feature propagation combined with cross-scale connections) is insufficient to build compound semantic relations for keypoint interpretation. Besides, the classical FPN multiscale regression layer is also not suitable for multiscale bboxes regression with irregular aspect ratios in anchor-free manners. In summary, an effective compound semantic description and more suitable regression layer designing can make the proposed MSFC-Net obtain a superior performance (e.g., 70.08% mAP and 73.45% mF1) on DIOR [99]. Then, from Table II line 1 to 8, we also can see anchor-based methods do not have much descending of mAP and mF1 comparing with the DOTA [98] dataset because of fixed anchor region classification and localization detection

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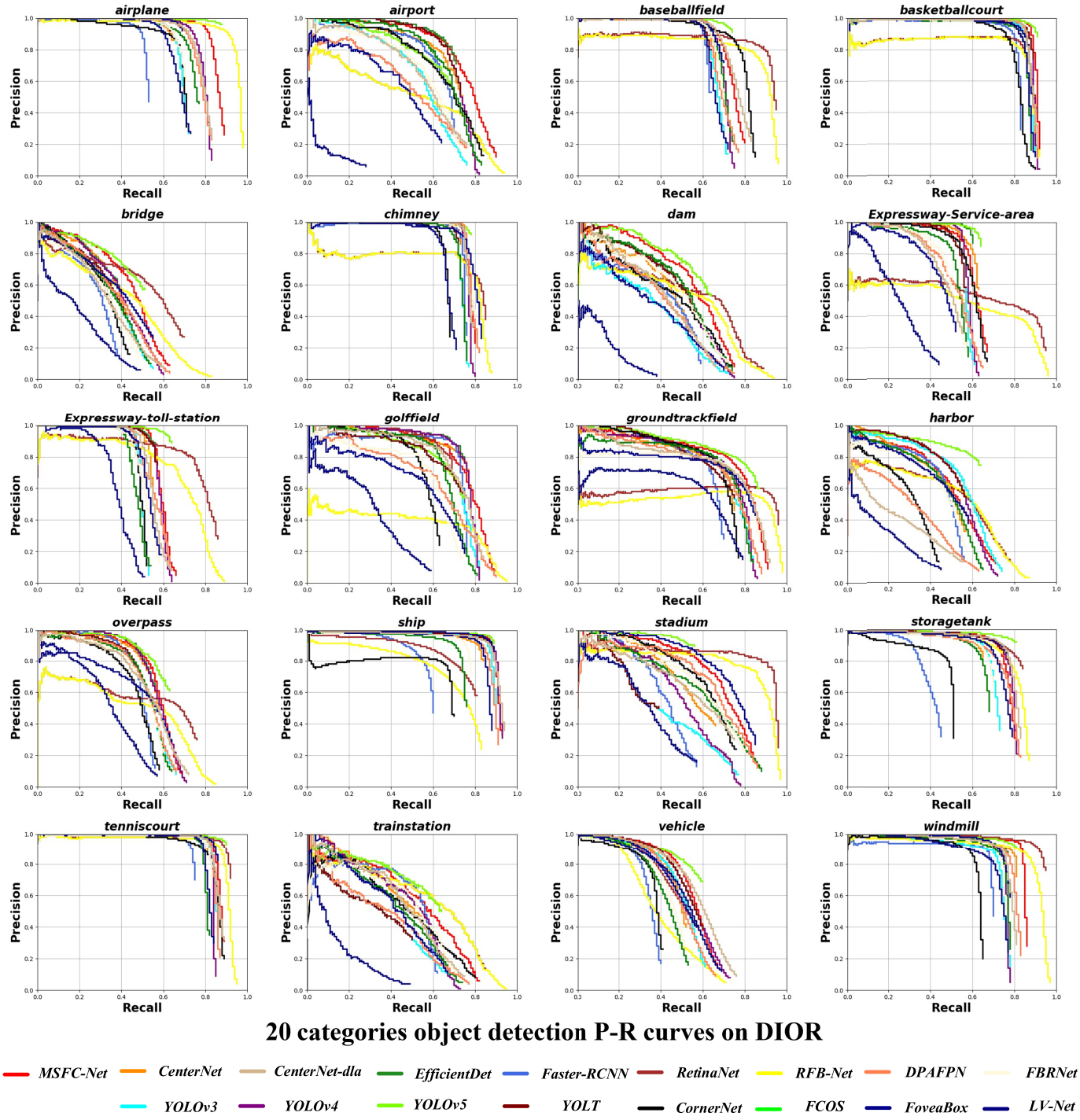


Fig. 10. P–R curves of the comparisons on DIOR [99].

mechanism, and then Yolo v4 [63] and v5 [64] still can perform good detection results. Nevertheless, in Tables I and II, these two strong versions of Yolo detectors also cannot outperform the proposed MSFC-Net on DOTA [98] and DIOR [99]. Finally, since our work mainly refers to EfficientDet [68] and CenterNet [69], then these detectors in Table II lines 8, 11, and 12 are chosen to compare with the proposed MSFC-Net. From these comparisons, we can see that the MSFC-Net can get much improvement than EfficientDet [68] and CenterNet [69].

As in the above analyses on DOTA [98] and DIOR [99], we can see that the proposed MSFC-Net is a powerful anchor-free one-stage detector for optical remote sensing object detection. In addition, several P–R curve comparisons and visualization detection results of the MSFC-Net on DOTA [98] and DIOR [99] are shown in Figs. 9–12. From these P–R curve comparisons, we can see that the proposed MSFC-Net achieves the competitive detection performance in all categories. As shown in Figs. 11 and 12, these visualization detection result shows that the MSFC-Net can adapt various optical remote sensing scenes containing multiscale, irregular aspect ratios and multiple categories. In addition, we also analyze the densely arranged object detection cases

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Fig. 11. Detection results of the MSFC-Net on DOTA [98].

TABLE V
SEMANTIC SEGMENTATION PERFORMANCE COMPARISONS ON GID

Method	*BG* (%)	*BU* (%)	*M* (%)	*W* (%)	*FL* (%)	*F* (%)	*PA* (%)	*mIoU* (%)
Deeplabv3+ [90]	71.7	**81.9**	**84.6**	85.3	74.2	**72.7**	75.3	62.8
GCN [89]	**77.7**	74.6	76.8	83.3	69.9	66.6	75.3	62.5
FCN [87]	76.0	77.4	81.8	82.5	69.3	64.1	74.6	59.8
CSFF (our)	75.7	76.0	80.1	**92.1**	**81.3**	67.7	**78.7**	**63.4**

with CornerNet [71], Faster-RCNN [59], Yolo-v5 [64], and the proposed MSFC-Net, and the keypoint prediction and detection results are shown in Fig. 13. From Fig. 13, we can see that object center points are more competitive than the corner points because there are many mismatches in the pairs of corner points in the CornerNet [71] detection results. Then, the results of Faster-RCNN [59] and Yolo-v5 [64] shown in Fig. 13(c) and (d) indicate that the anchor-based methods exhibit unstable detection results, which are influenced by predefined anchor regression process, IoU threshold setting, and NMS strategy.

D. Ablation Experiments

To further confirm the effectiveness of our designed modules in the MSFC-Net, several ablation experiments are carried out. First, as shown in Table III, several ablation experiments are

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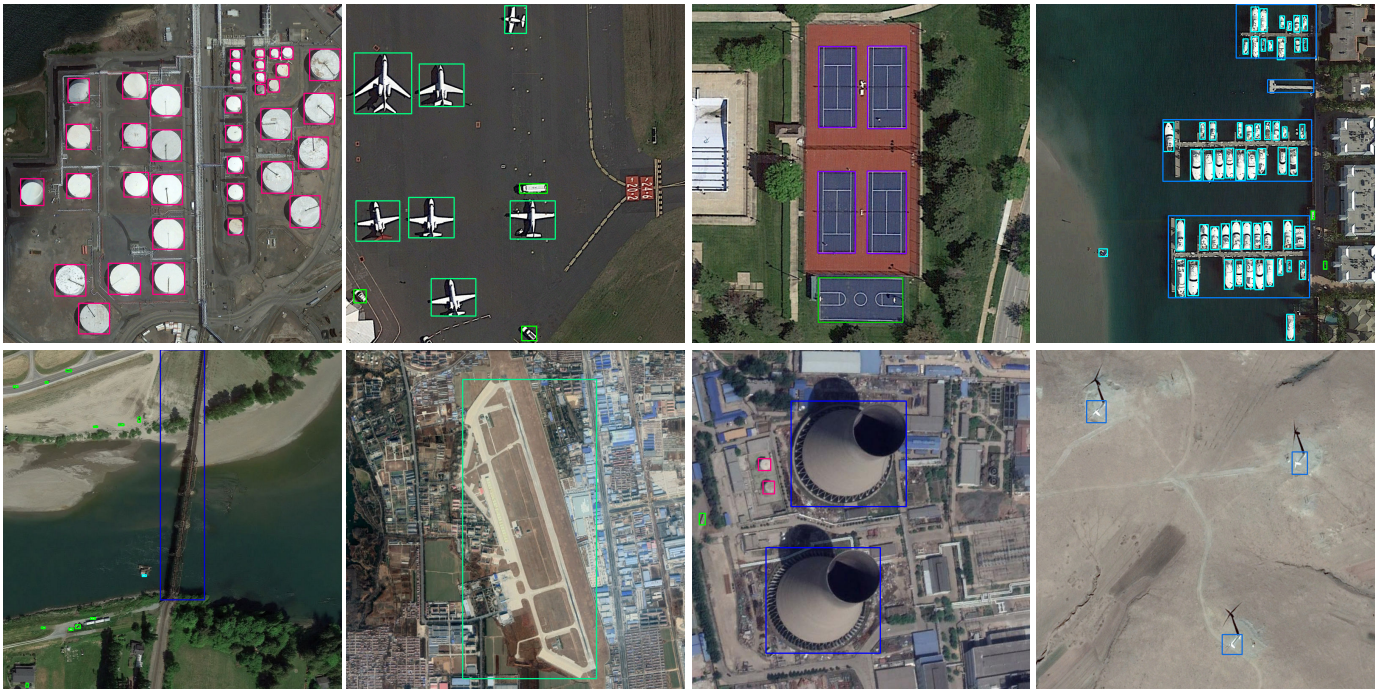


Fig. 12. Detection results of the MSFC-Net on DIOR [99].

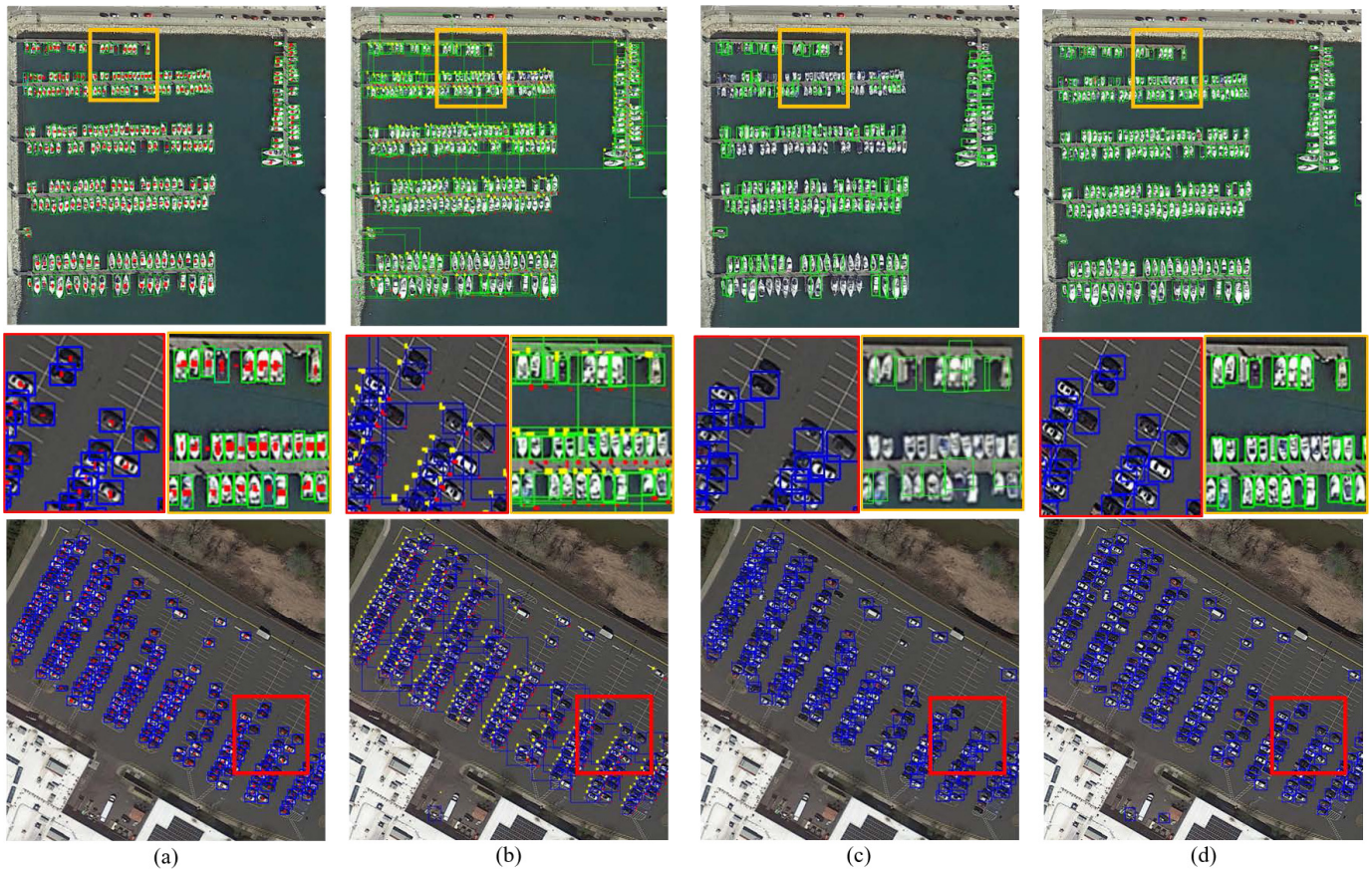


Fig. 13. Densely arranged object detection analysis. (a) Are the global and local ships and cars detection results of the MSFC-Net (e.g., the red pixels on objects are predicted center points). (b) Are the global and local ships and cars detection results of CornerNet [71] (e.g., the red and yellow pixels on objects are predicted pairs of corner points). (c) Are the global and local ships and cars detection results of Faster-RCNN [59]. (d) Are the global and local ships and cars detection results of Yolo-v5 [64].

carried out on DOTA [98], and CenterNet [69] is utilized as the baseline. In Table III line 1, we can see that the BiFPN can facilitate the multiscale RF propagation; thus, it can obtain a 1.01% mAP improvement over the baseline. Then, from

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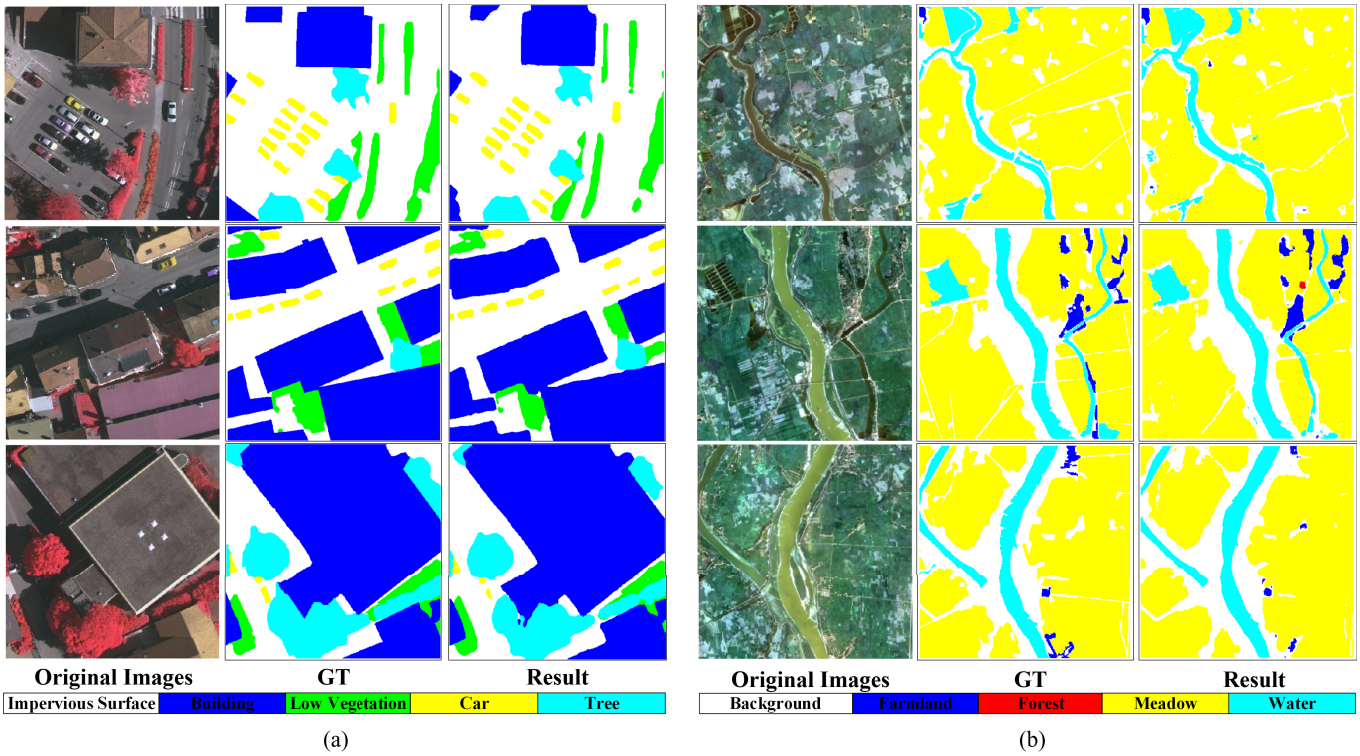


Fig. 14. Land use classification semantic segmentation results of the CSFF on (a) ISPRS [103] and (b) GID [104].

TABLE VI
RESULTS COMPARISON WITH MULTISCALE OR HIGH ASPECT RATIOS

Method	*AP (%)*			
	H	*HB*	*S*	*LV*
Yolo-v5 [64]	57.99	76.65	92.29	79.92
RetinaNet [66]	48.50	65.96	68.59	62.78
Our w/o FC layers	67.94	81.56	92.74	80.33
Our	**71.79**	**85.02**	**94.14**	**83.40**

TABLE VII
COMPARISON OF DIFFERENT BACKBONES

Backbone	*mAP#DOTA*	*Params(M)*	*GFLOPs*
VGG-19 [53]	77.31	23.4	185.1
DetNet-50 [57]	77.44	30.2	112.1
ResNet-50 [54]	77.65	26.9	104.3
CSPDarknet-53 [63]	77.84	30.3	109.7
EfficienNet-b0 [58]	72.85	**7.1**	**58.3**
ResNet-101 [54]	77.83	45.9	124.4
ResNeXt-101 [55]	78.07	90.1	169.6
ResNeSt-101 [56]	**80.26**	49.6	136.1

Table III lines 2, 3, and 4, the results indicate that the designed 3-D convolutions and CF operations applied in the BiFPN can facilitate effective compound semantic description to achieve accurate pixel-wise center point prediction, thereby yielding a 4.18% mAP improvement over the baseline. Next, to prove the effectiveness of the designed FC regression layers and hybrid loss function, as shown in Table III lines 5, 6, 7, and 8, we can see that compared with only the CSFF architecture applied on the baseline with multiscale FC regression layer, it can further obtain a 2.14% mAP improvement. In general, according to the ablation experiment results, we can see that the designed modules in the proposed MSFC-Net are very effective for optical remote sensing object detection.

1) CSFF Architecture: To demonstrate the pixel-level interpretation ability of CSFF module from optical remote sensing scenes, we take the pixel-level land use classification task into account as ablation experiments. Therefore, we employ several semantic segmentation networks (e.g., DeepLab-v3+ [90], GCN [89], FCN [87]) with different encoder-decoder ways as comparisons and carry out experiments on the ISPRS [103] and GID [104] datasets. These results are shown in Tables IV and V. We can see that the newly designed CSFF architecture can obtain the best performances of 87.3% PA and 70.9% mIoU on ISPRS [103] and 78.7% PA and 63.4% mIoU on GID [104]. Besides, several visualized pixel-wise semantic prediction results on ISPRS [103] and GID [104] are shown in Fig. 14, and these indicate that the designed CSFF can provide the accurate pixel-wise classification (e.g., the fine-boundary and uniform interior texture predictions) whether in a complex city or wide-area optical remote sensing scenes.

2) FC Regression Layers: To verify the validity of the designed FC regression layers, four object categories [e.g., LV, ship (S), HB, and helicopter (H)] with multiscale or high aspect ratio characteristics in DOTA [98] are chosen. We evaluate the detection effect by controlling whether to have FC regression layers in the MSFC-Net. The comparison results are shown in Table VI, where these anchor-based detectors of

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TABLE VIII
INFERENCE TIME AND PARAMETERS ANALYSIS OF DIFFERENT METHODS

Method	*Backbone*	*ImageSize*	*mAP#DOTA*	*mAP#DIOR*	*Params(M)*	*GFLOPs*	*IT (ms)*
Faster-RCNN [59]	ResNet-101 [54]	512×512	56.13	53.61	60.4	81.9	68
YOLOv5 [64]	CSPdarknet-53 [63]	512×512	74.22	68.57	44.4	83.0	**16**
RetinaNet [66]	ResNet-101 [54]	512×512	63.27	65.22	56.7	80.7	90
RFB-Net [67]	VGG-19 [53]	512×512	64.03	62.59	36.5	36.1	32
EfficientDet [68]	EfficientNet-b0 [58]	512×512	61.96	62.22	**7.7**	**25.0**	57
LV-Net [94]	ResNet-101 [54]	512×512	67.80	45.89	84.1	1246.0	115
FBR-Net [81]	ResNet-101 [54]	512×512	73.24	55.57	50.2	71.2	55
Centernet [69]	ResNet-101 [54]	512×512	73.94	63.86	53.6	75.1	42
Centernet-dla [69]	DLA-102 [69]	512×512	70.68	63.18	48.5	48.2	47
FCOS [70]	ResNet-101 [54]	512×512	76.95	61.17	50.9	70.77	54
FoveaBox [72]	ResNet-101 [54]	512×512	71.81	60.51	57.1	77.96	68
CornerNet [71]	Hourglass-54 [71]	512×512	63.33	56.66	111.5	294.1	129
DPAFPN [80]	ResNet-101 [54]	512×512	69.00	61.77	49.6	110.0	69
MSFC-Net	ResNeSt-101 [56]	800×800	79.21	**70.63**	46.8	126.9	60
	ResNeSt-101 [56]	512×512	**80.26**	70.08	46.8	126.9	59

Note: IT (ms) denotes Inference Time (ms)

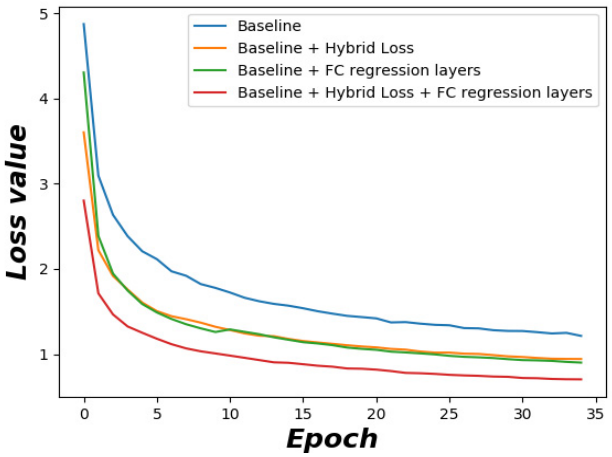


Fig. 15. Loss convergence analysis of the MSFC-Net.

RetinaNet [66] and Yolov5 [64] provide poor performances because their regression layers cannot regress the predefined multiple scales and aspect ratio anchors reasonably. In addition, in lines 3 and 4 of Table VI, the AP is increased by ∼3% for each chosen category; thus, the designed FC regression layers are very suitable for anchor-free detectors to deal with multiscale and irregular aspect ratio object detection. Finally, to verify that the designed FC regression layers combined with hybrid loss function can optimize the model convergence procedure, we evaluate loss convergence curves as shown in Fig. 15, obtained with different structure settings for the MSFC-Net. From Fig. 15, obviously, the red curve can rapidly converge on DOTA [98] when the baseline (i.e., MSFC-Net only has a CSFF module) introduces FC regression layers and hybrid loss.

3) Different Backbone Analysis: For proving the designed CSFF architecture can facilitate compound semantic description for pixel-wise center point interpretation and FC regression layers can boost multiscale bboxes regression with irregular aspect ratios, several widely used backbones (e.g., VGG-19 [53], ResNet-50 [54], DetNet-50 [57], CSPDarknet-53 [63], [64], EffecientNet-b0 [58], ResNet-101 [54], ResNeXt-101 [55], and ResNeSt-101 [56]) are employed for the proposed MSFC-Net. In Table VII, whether shallow layer or deep layer powerful backbone is utilized in the proposed MSFC-Net, it can obtain the competitive detection results with changeable and optional *Params* and giga floating-point operations (GFLOPs).

4) Inference Time and Parameters Analysis: Moreover, aside from accuracy, an object detector's inference speed, parameters, and complexity analysis are also important. Therefore, as shown in Table VIII, there are 13 famous detectors with different backbones as comparisons. Then, we prepare ten images, which are 512 × 512 or 800 × 800 pixels, to complete the inference speed test. Next, from Table VIII, we can see that our proposed MSFC-Net with ResNeSt-101 [56] backbone can provide superior performance with acceptable *Params*, *GFLOPs*, and *Inference Time* than other state-of-the-art detectors.

V. CONCLUSION

In this article, a powerful anchor-free one-stage detector called the MSFC-Net is proposed for optical remote sensing object detection, and it consists of CSFF architecture and FC regression layer. First, the CSFF architecture can leverage an effective compound semantic description, which can set up the pixel-wise coupled or uncoupled semantic relations for center point interpretation. Besides, CSFF architecture can be easily migrated into pixel-wise encoder-decoder process of land use classification, and it also can provide better performance. Second, a novel FC regression layer and its corresponding hybrid loss are first designed in an anchor-free manner, and it can reasonably adapt the multiscale bboxes regression with irregular aspect ratios without any priori information. Finally, extensive experiments are carried out on the DOTA [98] and DIOR [99] datasets. Then, the comparison results and detailed ablation experiments indicate that the proposed MSFC-Net can obtain superior performance than the state-of-the-art

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detectors. In addition, in future work, we will continuously exploit the more effective compound semantic description to further improve the performance of anchor-free detectors applied for optical remote sensing object detection. Next, the valid data augmentation and few-shot detection strategy will also be considered as the main emphasis of our research.

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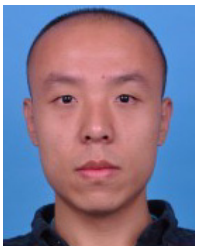
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